

AI-ASSISTED EMERGENCY CT TRIAGE

NeuroTriage AI

Prioritizing suspected intracranial hemorrhage for faster clinical review

MBA AI for Business | Management of AI products

Prototype — simulated/anonymized data — not for clinical use



The worklist can become the bottleneck

Emergency brain CTs compete for attention in a busy radiology workflow.

01 High volume

Multiple CT studies may arrive within a short period.

02 Unequal urgency

A critical finding may be buried among routine examinations.

03 Time-sensitive decisions

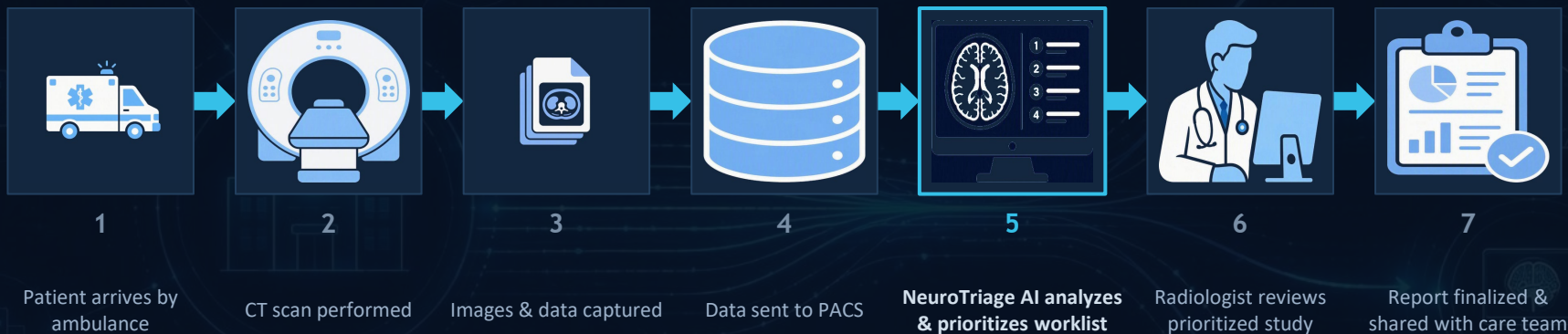
Suspected intracranial hemorrhage requires rapid recognition and review.



One critical study, buried in the queue

The opportunity: use AI to identify potentially critical studies early and bring them to the radiologist's attention sooner.

From patient arrival to diagnosis

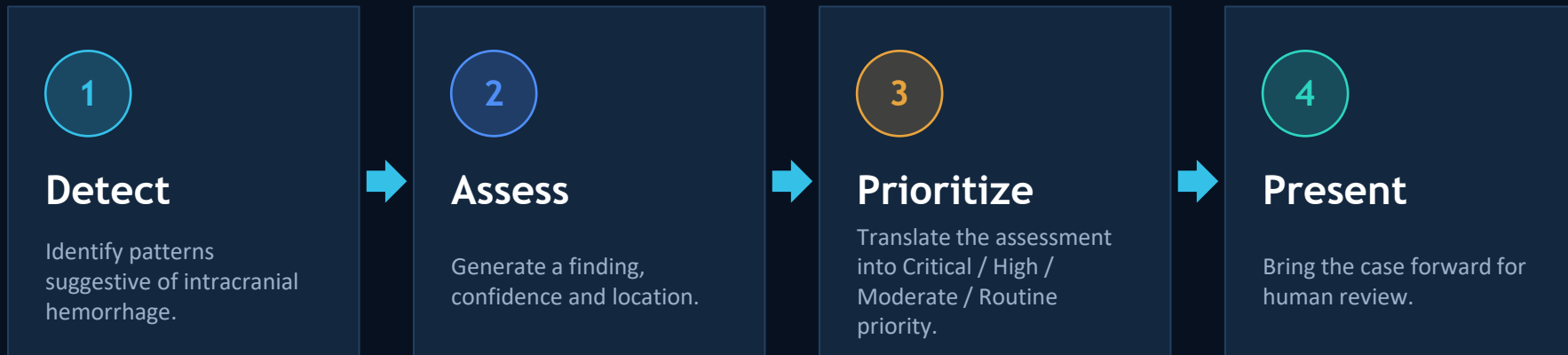


What is PACS?

Picture Archiving and Communication System — the hospital imaging system that stores, manages and delivers medical images to the right people at the right time.

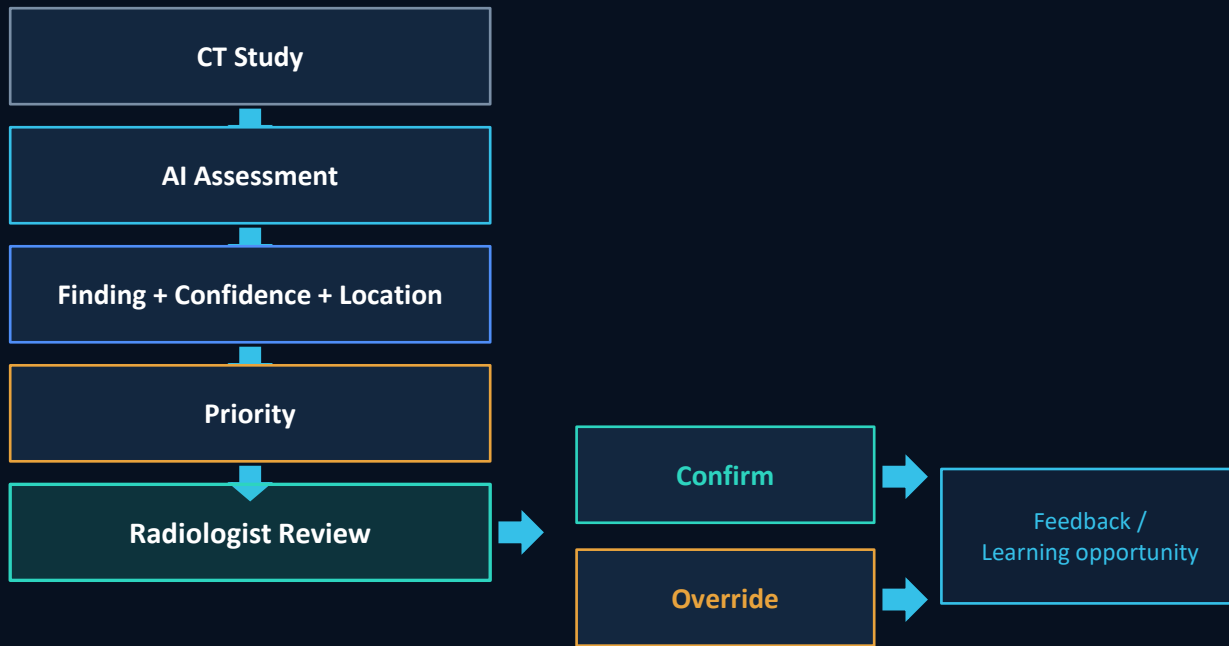
NeuroTriage AI sits inside this journey — it does not replace PACS or the radiologist, it surfaces potentially urgent studies sooner.

From image analysis to actionable prioritization



The product value is not simply detecting an abnormality — **it is helping the right case receive attention at the right time.**

AI supports a human-in-the-loop decision



Human remains accountable

- AI provides decision support
- Clinician reviews the evidence
- Clinician can override the AI
- Feedback is captured

A working MVP — not just a concept

Realistic data layer

- Anonymized DICOM CT studies
- 6 real study datasets
- 4 ICH-positive cases
- 2 normal cases

AI-assisted workflow

- Finding assessment
- Confidence score
- Priority classification
- Hemorrhage location
- Automatic worklist reprioritization

Reading experience

- CT slice navigation
- AI overlay
- Jump to Finding
- Brain / Blood-ICH windowing
- Confirm / Override

The prototype demonstrates the end-to-end product workflow, from CT study arrival to clinician decision.

AI inference is simulated for this prototype; the workflow is designed to support a future validated AI inference service.

Critical cases rise to the top

CT Brain Worklist · prototype screen

CT Brain Worklist
5 studies

Reset Demo

Simulate New CT Study

PATIENT NAME	STUDY	STUDY STATUS	AI STATUS	AI PRIORITY
Jordan Ellis	CT Brain W/O Contrast	Completed	Ready	Critical
Wei Zheng	CT Brain W/O Contrast	Completed	Ready	Critical
Priya Nair	CT Brain W/O Contrast	Completed	Ready	High
Ahmed Farouk	CT Brain W/O Contrast	Completed	Ready	Moderate
Maria Castillo	CT Brain W/O Contrast	Completed	Ready	Routine

Critical → immediate attention

High → elevated priority

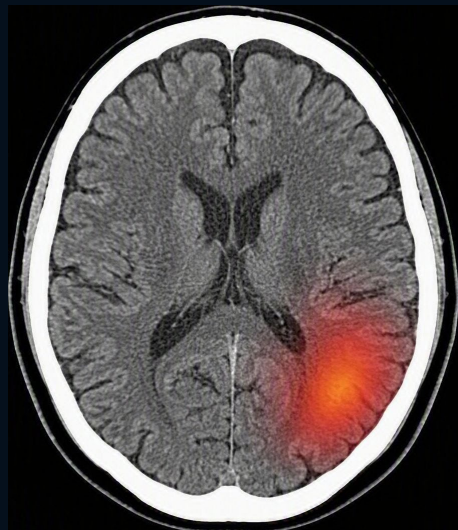
Moderate → routine escalation

Routine → standard workflow

Simulate New CT Study → the worklist
reorders automatically

As studies become available, the worklist automatically reprioritizes based on the AI assessment. Screen shown is the working prototype with simulated, anonymized studies.

From “something is urgent” to “show me where”



1. AI Assessment

Suspected ICH · 93% confidence

2. Explainable location

Left parietal lobe

3. Jump to Finding

One click takes the reviewer directly to the highlighted slice.

Brain ↔ Blood/ICH windowing allows the reviewer to inspect the finding under different display settings.

AI recommends. The clinician decides.

AI

- Detection
- Confidence
- Prioritization
- Visual indication

HUMAN

- Review
- Clinical context
- Confirmation
- Override

SYSTEM

- Audit and feedback
- Transparent workflow
- Explicit uncertainty

The prototype deliberately avoids presenting AI output as a final diagnosis.

What value could NeuroTriage AI create?

Patient

Potentially faster identification of urgent cases.

Clinician

Reduced effort in finding high-priority examinations.

Hospital

More efficient use of radiology capacity.

Business

Potential platform for AI-assisted workflow products.

Potential pilot KPIs

- Time from study completion → radiologist review
- Time to review of suspected ICH
- Sensitivity / false-negative rate
- False-positive rate
- Override rate
- AI adoption / usage
- Impact on radiologist workflow

These are proposed success measures — they were not validated by this prototype.

A successful prototype is not a clinical validation

Demonstrated

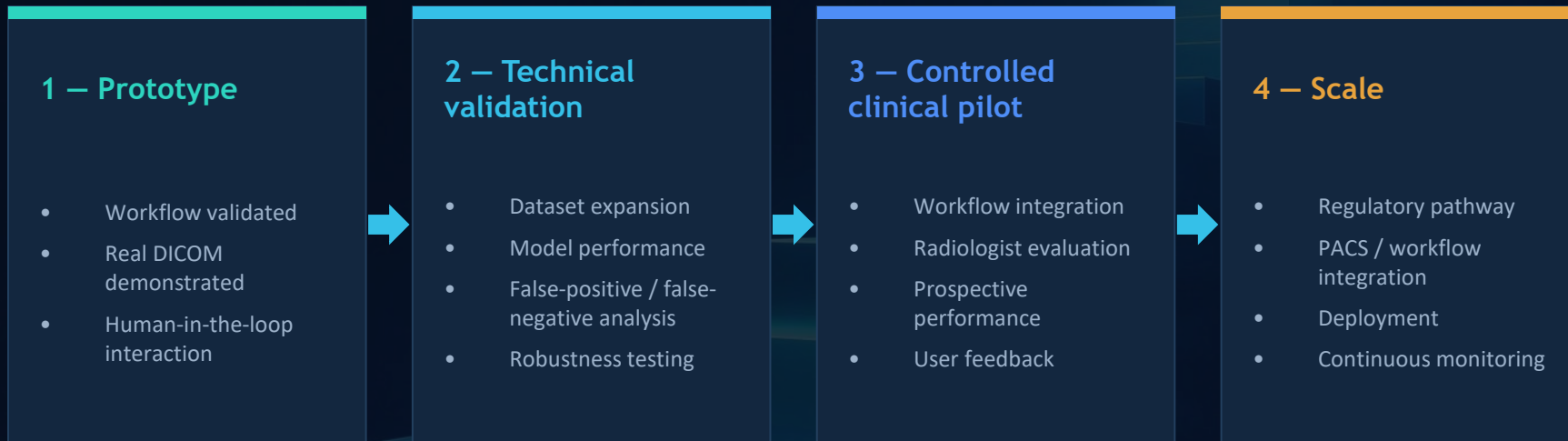
- End-to-end workflow
- Real anonymized CT DICOM integration
- Window/level handling
- AI-style prioritization
- Human confirmation / override
- Worklist reprioritization
- Explainable visual indication

Not yet demonstrated

- Clinical diagnostic performance
- Generalization across populations and scanners
- Prospective clinical benefit
- Regulatory clearance
- Production-scale integration
- Patient outcome improvement

The prototype validates the product concept and workflow — not clinical efficacy.

From prototype to controlled pilot



Recommendation: proceed to controlled validation, with clinical performance, workflow impact, safety and regulatory requirements treated as explicit stage gates.